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Effect of Surface Roughness on Film Cooling Performance

An experimental investigation of film cooling on a rough surface has been performed using a mass transfer technique. Two injection geometries (one and two rows of holes) and six roughness patterns were used. The density of the injected gas was close to that of the free-stream air. The presence of roughness causes a decrease in the spanwise-averaged effectiveness for both injection geometries at low blowing rates, and an increase in the effectiveness for one-row injection at high blowing rates, but not for two-row injection. The results for two-row injection (with or without roughness) can be correlated by a parameter used to correlate slot-injection data. The influence of roughness pattern on the average effectiveness and on the lateral uniformity of the effectiveness is described.

Introduction

Film cooling has been widely used as a means of protecting solid surfaces exposed to high-temperature gas streams, especially in gas turbines. Past analytical and experimental studies, in general, assume a smooth wall surface. However, in the operation of gas turbines, the surface often becomes rough either by corrosion or by deposits from contaminants in the fuel or brought in with the atmospheric air. The effect of roughness on film cooling performance is thus of practical importance.

The present paper reports the results of film cooling measurements with discrete hole injection of air into an air stream flowing over a rough surface. A mass transfer technique is used to obtain the analogue of the adiabatic wall temperature. Helium is used as tracer gas in the secondary (injected) flow, and its concentration is determined by gas chromatography. In the reported results, the density of the injected gas is close to that of the mainstream air.

2 Apparatus

Measurements are taken in an open cycle wind tunnel with a 25cm-wide by 13cm-high test section. Mainstream air flows through a filter, a flow straightener, a series of three screens, and a 9-to-1 contraction section before entering the test section. After the test section, the air passes through a blower which discharges it into the atmosphere.

The injection system includes an orifice-metered air blower system to supply the air. Helium comes from a high-pressure cylinder and is regulated by capillaries. The air and helium flow through a mixing chamber and a plenum chamber before being injected into the mainstream. The air-helium mixture density is kept at 0.95 of the mainstream air density. A separate plate (removable) into which the jet holes are drilled is part of the test surface. It measures 24cm by 8cm (in flow direction) and sits on a 1cm step around its edges so that its lower open surface is 22cm by 6cm. The thickness of the plate is 1.9 cm.

The velocity of the main flow in the test section is either 30 or 45 m/s. The boundary layer on the test surface is "tripped" by a wire at the outlet of the contraction section; it is both turbulent and uniform across the width of the section. The displacement thickness at $X/D=0.0$ (measured in the absence of the injection holes) is 1.5 mm when the free-stream velocity is 45 m/s.

Two injection geometries are used – namely, one or two rows of injection holes. The injection holes are circular, $D=6.35$mm, and inclined by 35 deg to the mainstream flow. Eleven holes are arranged at three diameter centerline to centerline spacing across the span for the one-row case. For the two-row geometry, the staggered second row of ten holes is situated 2.6D upstream of the first row (Fig. 1).

Gas samples are taken through suction taps installed at specific locations along the test surface. A vacuum pump draws the gas through the taps. The gas flows through vinyl tubes to glass sampling loops which store the gas samples for later analysis in a gas chromatograph. The sampling process and the gas analysis procedure are similar to that used in reference [1]. Twenty-eight taps are arranged in four columns. The lateral positions of each of the four columns (e.g., Taps A-D, Fig. 1) are $Z/D=0.0$, 0.5, 1.0, and 1.5, respectively. The streamwise positions of the seven taps within each column are $X/D=4.7$, 8.7, 14.7, 20.7, 28.7, 38.7, and 50.7, respectively, for one-row injection, and are all smaller by $X/D = 0.7$ for two row injection, e.g., 4.0 instead of 4.7.

The surface roughness geometry was selected to simulate the roughness encountered on turbine blades. A blade with film cooling on both the suction and pressure sides was inspected for this purpose. The blade came from a turbine used in a relatively dirty environment. On the suction side,

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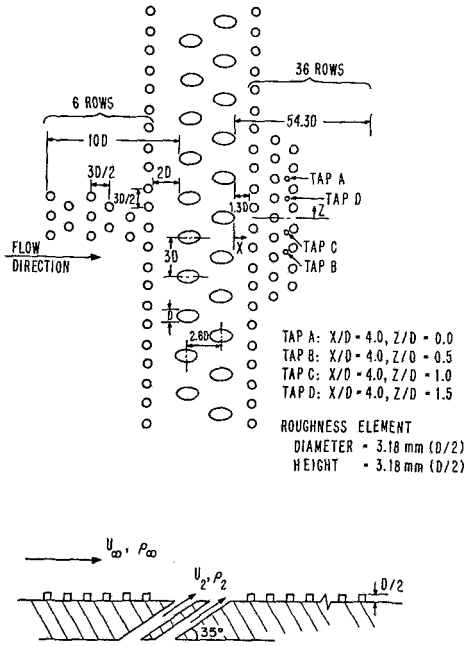

Fig. 1 Schematic of test plate for one roughness pattern (BH2H)

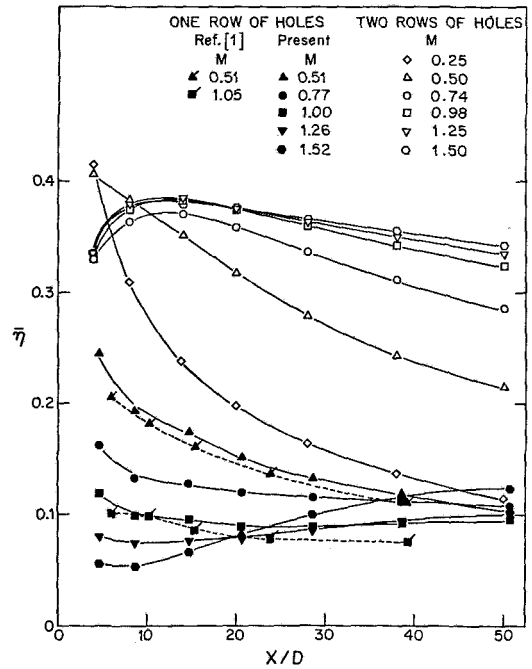

Fig. 2 Spanwise-averaged effectiveness as function of downstream position for smooth surface

roughenss was present upstream of the film cooling holes while, on the pressure side, there was roughness mainly downstream of the holes with some accumulation also occuring upstream. The height and spacing of the surface roughness were measured. A regular geometrically well-defined roughness pattern was designed to simulate the pattern found on the turbine blade.

Surface roughness in the tests is provided by cylindrical roughness elements with a diameter of 3.2mm ($D/2$) and a height of either 3.2mm ($D/2$) or 1.6mm ($D/4$) glued onto the test surface in a regular array. The roughness elements are in staggered rows with 1.5 D spacing between the rows and between individual elements in a row. Each row of roughness elements essentially covers the width of the tunnel. The height of the taller roughness elements ($D/2$) is modelled from the measurements of the inspected turbine blade section.

Six different roughness patterns are used – a combination of one of the two different-sized roughness elements placed in one of three different locations relative to the injection holes: roughness upstream only; downstream only; and both upstream and downstream. The upstream roughness has six rows of elements. The last row is 2D upstream of the upstream edge of the holes (the first row of holes in the two-row injection case). The downstream roughness is provided by 36 rows of elements, the first row is 2D downstream of the downstream edge of the injection holes (1.3D from the downstream edge of the second row of holes in the two-row case). A schematic of one of the six roughness patterns used in the present study is shown in Fig. 1. It is designated BH2H roughness. The notation of the roughness patterns is described in Appendix A.

3 Results

3.1 Effectiveness on a Smooth Surface. The film cooling effectiveness for a smooth surface was measured in the apparatus to provide a reference for the effect of surface roughness. Figure 2 shows the spanwise-averaged effectiveness, $\bar{\eta}$, as a function of the dimensionless downstream distance, X/D, for both one and two rows of injection holes. Some results from reference [1] are shown for comparison.

For two-dimensional film cooling (no variation of mainstream or injected flow across span), a film cooling parameter permits good correlation of the effectiveness data [2]. For the

Nomenclature

D = diameter of injection hole
M = blowing rate, $M = \rho_2 U_2/\rho_\infty U_\infty$
Re = Reynolds number
U = mean velocity
X = distance downstream of trailing edge of downstream hole
Z = lateral position from centerline of downstream hole
η = local film cooling effectiveness
η_{max} = maximum value of η at a given Z
η_{min} = minimum value of η at a given Z
$\bar{\eta}$ = laterally-averaged film cooling effectiveness
ξ = a dimensionless parameter used in correlating two-dimensional film cooling data; defined in equations (1) and (2)
ζ = lateral smoothness parameter, $\zeta = (\eta_{max} - \eta_{min})/\bar{\eta}$ at a given X/D
μ = dynamic viscosity
ρ = density
$<\bar{\eta}>$ = area-averaged effectiveness, obtained by streamwise averaging of $\bar{\eta}$ between $4 \leq X/D \leq 50$
$<\bar{\eta}/\bar{\eta}_{smooth}>$ = streamwise averaging of $\bar{\eta}/\bar{\eta}_{smooth}$ between $4 \leq X/D \leq 50$ ($\bar{\eta}_{smooth}$ is $\bar{\eta}_{NO1H}$ or $\bar{\eta}_{NO2H}$ for one-row or two-row injection, respectively)

Subscripts

2 = secondary or injection flow
smooth = no surface roughness
∞ = mainstream flow

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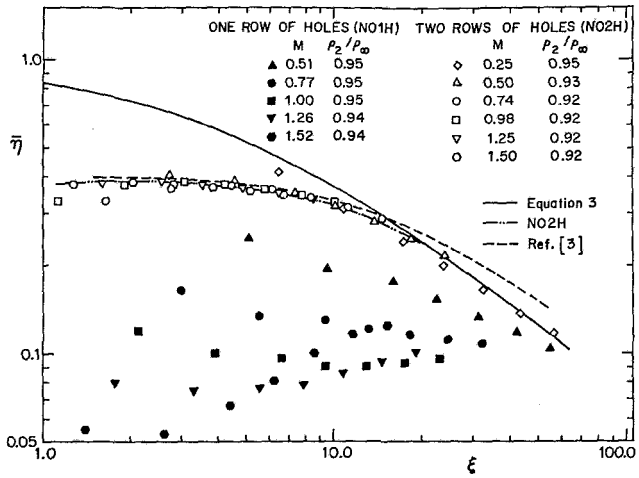

Fig. 3 Variation of spanwise-averaged effectiveness, $\bar{\eta}$, with film cooling parameter, ξ, for smooth surface

present study, this two dimensional parameter, ξ, is defined to be

$$\xi = \left(\frac{X}{D} + 1.909\right) \frac{6}{\pi M} \left(\frac{\mu_2}{\mu_\infty} \frac{\pi}{6} \mathrm{Re}_2\right)^{-1/4} \tag{1}$$

for two rows of injection holes [3], and

$$\xi = \frac{X}{D} \frac{12}{\pi M} \left(\frac{\mu_2}{\mu_\infty} \frac{\pi}{12} \mathrm{Re}_2\right)^{-1/4} \tag{2}$$

for one row of holes, where, in both cases, $\mathrm{Re}_2 = \rho_2 U_2 D/\mu_2$.

The variation of $\bar{\eta}$ with ξ for the present results is plotted in Fig. 3. Also plotted as a solid line on Fig. 3 is a theoretical prediction of two-dimensional film cooling adapted [2] from an analysis by Kutateladze and Leont'ev:

$$\eta = (1 + 0.249\ \xi)^{-0.8} \tag{3}$$

and a dashed line representing results presented in reference [3] for injection through two rows of holes with a smooth plane surface. A curve (NO2H) is also fitted to the two-row smooth surface data of the present study to be used in later comparisons.

From Fig. 3 one can see that the parameter ξ correlates film-cooling results for two rows of injection holes but does not correlate the data for one row of holes. With the two-row injection geometry, the interaction between the jets from the two staggered rows reduces the lateral nonuniformity. This interaction prevents the mainstream from flowing around and under individual jets and the rows of jets behave like a pseudo-two-dimensional sheet to the main flow. Even so at high blowing rates and/or small x (small ξ), the effectiveness is lower than that found with two-dimensional film cooling; the effectiveness at low blowing rate with the two staggered rows is comparable to that with a two-dimensional slot.

A characteristic of the discrete-hole injection geometry is the three-dimensional nature of the flow field and the resulting lateral nonuniformity in the effectiveness. A dimensionless lateral smoothness parameter, ζ, is defined at a given X/D by

$$\zeta = \frac{\eta_{\max} - \eta_{\min}}{\bar{\eta}} \tag{4}$$

The variation of ζ with X/D is presented in Fig. 4 for one and two-row injection geometries. The smaller value of ζ for the two-row geometry indicates a more uniform lateral distribution of effectiveness.

3.2 Influence of Roughness on the Average Effectiveness. Quantitative effects of the different roughness sizes and locations are investigated by comparing the effectiveness on

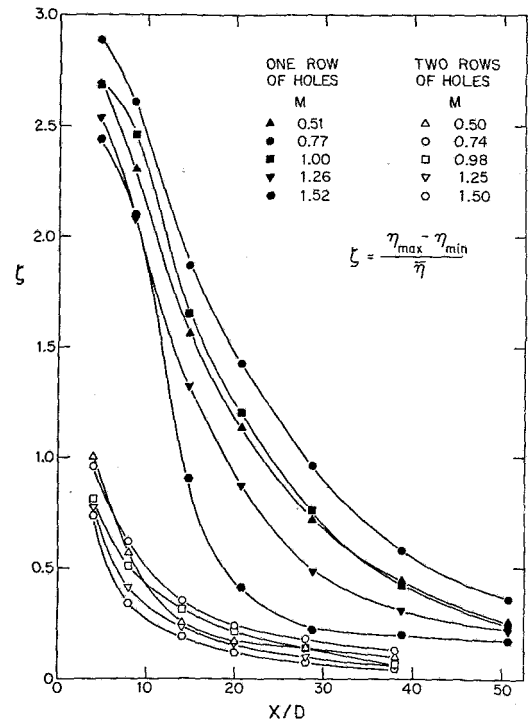

Fig. 4 Variation of lateral smoothness of effectiveness parameter, ζ, with downstream position for smooth surface

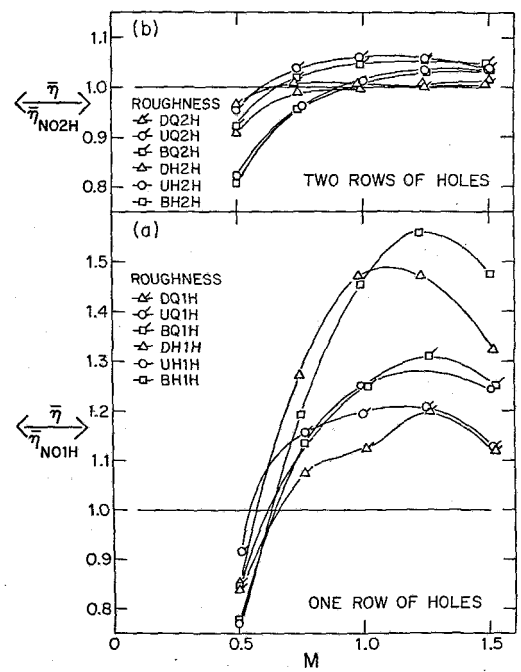

Fig. 5 Effect of roughness (relative to no roughness results) on area-averaged effectiveness as a function of blowing rate.

the rough surface to that on the smooth surface for the same injected flow. In general, the presence of roughness will affect the average effectiveness, the lateral distribution of the local effectiveness, the wall shear stress, and the heat transfer. The present investigation does not provide data for the examination of the last two factors. This section examines the effect of roughness on the average effectiveness. The effect of roughness on lateral variation of η is described in the next section.

The ratio of the average effectiveness with roughness to that with the smooth surface was calculated from the results

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of the measurements. The variation of $\bar{\eta}/\bar{\eta}_{smooth}$ is relatively small, less than 30 percent over the entire range of X/D other than for the BH1H roughness at $M > 1.0$, which has a variation in $\bar{\eta}/\bar{\eta}_{smooth}$ of about 50 percent over the investigated range of X/D. This ratio ($\bar{\eta}/\bar{\eta}_{smooth}$) was averaged over the entire range of X/D measured ($4 \leq X/D \leq 50$). The resulting average quantity, $<\bar{\eta}/\bar{\eta}_{smooth}>$, is plotted as a function of the blowing rate, M, for different roughnesses in Figs. 5(*a*) and 5(*b*) for the two injection geometries. The presence of roughness causes a decrease in the effectiveness over that of the no-roughness runs for all roughness patterns at small blowing rates. At larger M, roughness can result in a significant improvement of the effectiveness for one row of injection holes, while for two rows of injection holes the improvement is minimal. These trends can be explained in the following way. For small M, the jet momentum is small and the injected flow remains attached to the wall. The presence of roughness induces more vigorous mixing between the mainstream and the secondary flows near the wall, decreasing the effectiveness due to greater dilution of the coolant. The situation is more complex at larger blowing rates. In the case of one row of holes, the high momentum of the secondary jets causes "blow-off" and the jets penetrate into the main flow away from the wall. The roughness causes more turbulent mixing and reduces the penetration of the secondary jets, thus improving the effectiveness. Note that the transition from a decreased to an increased effectiveness with roughness occurs at $M \sim 0.6$ for one row of holes; this roughly corresponds to the "blow-off" blowing rate for the density ratio, ρ_2/ρ_∞, used. With two rows of injection holes there is little improvement at higher M when roughness is present as the interaction between the staggered rows of jets already provides an almost continuous flow across the span, preventing "blow-off"; the effect of additional mixing caused by the presence of roughness is small.

With injection through a single row of holes, the effectiveness for all roughness patterns relative to that for no-roughness increases as M increases, reaches a maximum around $M = 1.25$, and decreases as M increases further. This maximum is primarily due to the minimum—around $M = 1.25$ (cf, Fig. 2 or 3)—of the area-averaged effectiveness, $<\bar{\eta}>$, for a smooth surface. The decrease in $<\bar{\eta}>$ with increasing M for a smooth surface in the range of $0.5 < M < 1.25$ is due to the "blow-off" of jets from the surface and the subsequent increase with M is due to the reattachment of the jets back to the wall by deflection from the main flow.

The effect of roughness element height ($D/2$ or $D/4$) can be observed on Fig. 5. For one row of holes, on a configuration-to-configuration basis, the taller roughness elements produce a bigger decrease in effectiveness relative to that of no-roughness case at $M < 0.6$ and generate a bigger improvement at $M > 0.6$ as compared to the shorter roughness elements. With the two-row geometry a similar influence is found (see Fig. 5(*b*)), at least at small M. Compared to the shorter roughness elements, the taller roughness elements cause greater mixing of the jets near the wall; at low M, this decreases the area-averaged effectiveness; at higher M, it will reduce the "blow-off" effect (for one-row-of-holes) and increase the area-averaged effectiveness.

The influence of roughness location (roughness downstream only "*D*," upstream only "*U*," and both upstream and downstream "*B*") can also be observed on Fig. 5. For the one-row geometry, the "*B*" configuration provides the largest improvement at large M, while the "*U*" configuration yields the smallest improvement. For moderate blowing rates, say $0.6 < M < 1.2$, the "*B*" configuration has a better improvement than the "*U*" configuration. However, DH1H now has the biggest improvement and DQ1H has the smallest improvement. At low M ($M < 0.6$), there is insufficient data to show any clear-cut effect of roughness location.

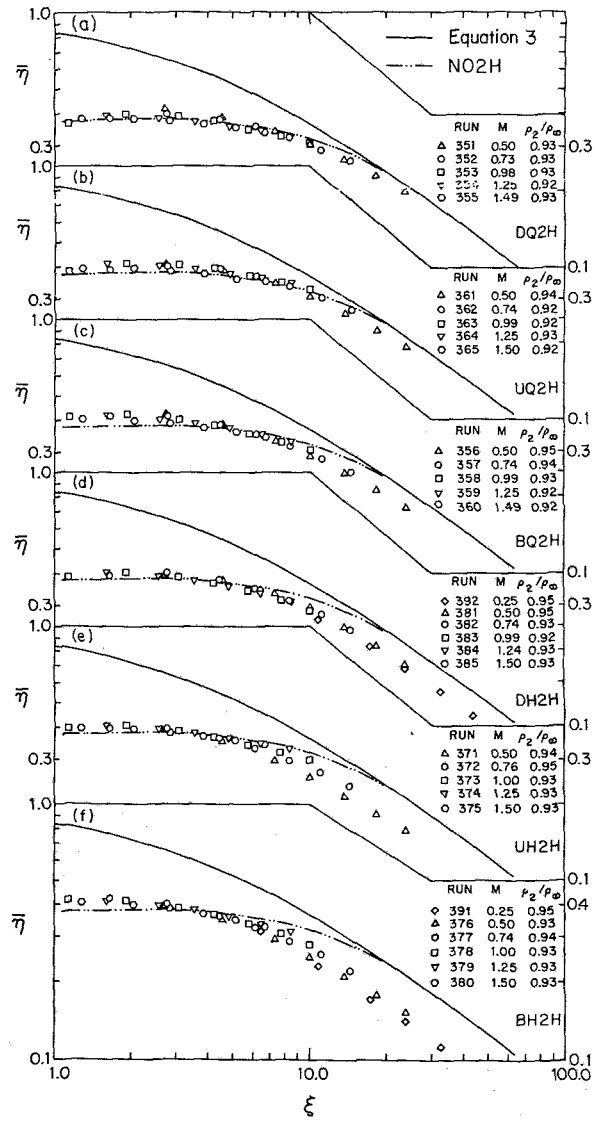

Fig. 6 Variation of spanwise-averaged effectiveness, $\bar{\eta}$, with film cooling parameter, ξ, for various roughness configuration with injection through two rows of holes.

Upstream roughness will increase the turbulent intensity in the approaching boundary layer, and downstream roughness will increase the mixing between the jets and the main flow. The effect of upstream roughness is less pronounced at large M where turbulent mixing is already vigorous; however, for the larger roughness height located upstream at small M, enough turbulence will be generated to have a big negative effect on the area-averaged effectiveness. The downstream roughness has a small effect at smaller M and produces different behaviors at larger M depending on the roughness sizes. Before the blown-off jets reattach to the wall ($0.6 < M < 1.2$), DH1H roughness produces the largest effectiveness and DQ1H the smallest; after reattachment ($M > 1.2$), both DH1H and DQ1H have an effect between that of the corresponding "*U*" and "*B*" configurations.

The above observations suggest that mixing between the jets and the mainflow (increased by downstream roughness) plays a more important role in reducing the "blow-off" phenomenon than approaching turbulent intensity (increased by upstream roughness) for the larger roughness size. For $0.6 < M < 1.2$, where the penetration of the jets is still small, having upstream roughness in addition to downstream roughness actually has an adverse effect—i.e., the im-

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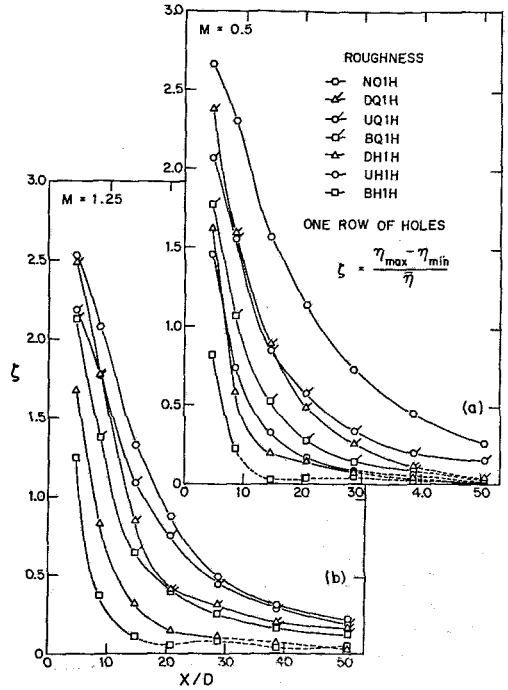

Fig. 7 Variation of lateral smoothness of effectiveness parameter, ζ, with downstream position for one-row injection with roughness

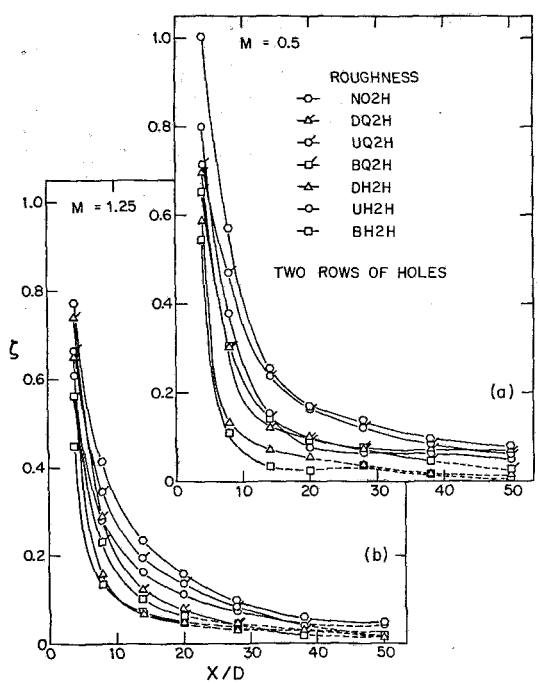

Fig. 8 Variation of lateral smoothness of effectiveness parameter, ζ, with downstream position for two-row injection with roughness.

provement in effectiveness is slightly less (BH1H vs DH1H). For $M > 1.2$, the jet momentum is high enough that both upstream and downstream roughnesses are needed for best reduction of "blow-off". With the shorter roughness elements, both the approaching turbulence level and the downstream mixing of jets and mainflow are moderate. In this case, having roughness elements upstream has a bigger effect than having them downstream for $0.6 < M < 1.2$; at $M > 1.2$, their individual effects are comparable.

With two-row injection, the effect of roughness can be investigated by using the parameter ξ. The variation of $\bar{\eta}$ with ξ for runs with two rows of holes is presented in Fig. 6. Figures 6a-f give results for DQ2H, UQ2H, BQ2H, DH2H, UH2H, and BH2H roughnesses, respectively. Also plotted on each of the figures is a curve which approximates the smooth surface (NO2H) results from Fig. 3. All the different roughness patterns can be correlated by ξ, suggesting a pseudo-two-dimensional behavior. The effect of roughness on effectiveness is minimal for $\xi < 6.0$, though, for some roughness patterns, there is a slight improvement over the no-roughness results. Small ξ corresponds to situations of high M and/or low X/D. For $\xi > 6.0$, UH2H and BH2H roughnesses (Figs. 6(*e*) and 6(*f*)) cause the biggest decrease in effectiveness; DQ2H and UQ2H (Figs. 6(*a*) and 6(*b*)) cause a small drop in effectiveness and, with BQ2H and DH2H (Figs. 6(*c*) and 6(*d*)), the decrease in effectiveness is moderate. The decrease in effectiveness is due to greater mixing of the injected flow with the mainstream when the roughness is present.

3.3 Effect of Roughness on Lateral Smoothness. The effect of roughness on the variation of η across the span can be assessed by comparison of ζ for various roughness types, as shown in Fig. 7 and 8 for one-row injection and two-row injection, respectively. Dotted lines indicate situations where the wall concentration profile can practically be considered as constant in the Z direction.

The presence of roughness results in a flatter profile as compared to the no-roughness case. As expected, the relative change is more pronounced for one row of holes since the wall concentration profile for NO2H is already relatively smooth. More mixing due to the presence of roughness causes the increased uniformity of the local effectiveness over the span of the surface. The taller roughness elements induce greater mixing and generate a flatter lateral distribution than the shorter roughness elements except perhaps for upstream roughness and two-row injection at small M and $X/D < 5.0$ (see Fig. 8(*a*)).

For one row of holes the "*B*" configuration gives the most uniform profile. The "*D*" and "*U*" configurations produce comparable results with the "*D*" configuration giving a slightly flatter lateral profile at large X/D, and the "*U*" configuration causing a somewhat more uniform distribution for small X/D. For two rows of holes the effectiveness is essentially uniform at X/D over 20.0; significant variations of effectiveness with Z occur for $X/D < 10.0$. Here the "*B*" configuration gives the flattest profile, with the "*D*" configuration next, and then the "*U*" configuration. The *UQ2H* configuration has almost the same lateral smoothness in effectiveness as does a smooth surface.

4 Conclusion

Surface roughness typical of that found on gas turbine blades exposed in a system operating in a relatively dirty environment can have a significant impact on film cooling performance – particularly with injection through a single row of holes. At low blowing rates, there is a decrease in the adiabatic wall effectiveness of the order of 10-to-20 percent over that compared to a smooth surface under similar conditions. However, at higher blowing rates, when the jets tend to lift-off from the surface, the film cooling performance is improved significantly, by as much as 40-to-50 percent under some conditions. This improvement follows from the greater turbulence and mixing with the rough surface, which tends to dissipate the injected flow and prevent it from penetrating as far into the mainstream. The effects are most pronounced with the largest roughness elements and generally also when the roughness is both upstream and downstream of the injection section. Although the general trends are similar with two rows of injection holes, the net effect is much smaller, particularly relative to the improvement that occurs at large blowing rates.

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For injection through a single row of holes, a rough surface also produces a relatively uniform lateral distribution of film cooling effectiveness, as compared to that of a smooth surface. Similar trends were found with two rows of injection holes, although this injection system does not have a large lateral variation, even with a smooth surface. Lateral variation of effectiveness could be important in affecting temperature gradients and thermal stresses.

Care should be taken in interpretating the significant improvements and the magnitude of the effectiveness as well as the lateral smoothness of the effectiveness in terms of getting better film cooling performance with a rough surface. With such a surface, the magnitude of the heat transfer coefficient is expected to be larger than that of a smooth surface with or without film cooling. This would also, perhaps, point to the greater need of having film cooling when the surface is rough.

Acknowledgment

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APPENDIX

Classification of the Test System

In addition to the flow rates, parameters that were varied in the present test relate to the roughness pattern, the flow geometry, and the injected gas. To classify each series of tests, a four-character word is employed. This includes the following parameters:

1. roughness location;
2. type of roughness elements;
3. number of rows of holes; and
4. density ratio.

The first character designates the location of roughness used:

U = roughness elements upstream of injection hole only
D = downstream only
B = both upstream and downstream
N = no roughness

The second character designates the type of roughness elements employed (all elements had a diameter half that of the injection holes):

H = 3.2 mm height, equal to half the injection hole diameter (or $D/2$)
Q = 1.6 mm height, equal to one-quarter the injection hole diameter (or $D/4$)
0 = no roughness

The third character designates the injection geometry:

1 = one row of injection holes
2 = two rows of injection holes

The last character designates the injection gas density ratio:

H = density ratio 0.95, helium as tracer gas

EXAMPLE: Roughness type BH2H means density ratio 0.95, two rows of holes, roughness element of height D/2 placed both upstream and downstream of injection holes (see Fig. 1).

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